

CASE STUDY

ZEAL HOTEL

Exeter Science Park, Exeter

MAIN CONTRACTOR
RED Construction

PROJECT
New Build, Hotel

PACKAGE
Roofing, waterproofing & insulation · timber & solar PV cladding

UK'S FIRST BRANDED NET-ZERO CARBON HOTEL



- **Bristol based - Nationwide Coverage**
- **Full envelope contractor**
- **Roofs built to last. Guaranteed.**

PROJECT OVERVIEW

The UK's first branded net-zero carbon hotel

Zeal Hotel at Exeter Science Park is the UK's first branded net-zero carbon hotel, marking a major milestone in sustainable development and hospitality, delivered by main contractor RED Construction.

Designed by ECE Westworks Architects, the four-storey, 142-bedroom hotel brings together sustainability and innovative design. The Worx Group delivered the roofing, waterproofing and insulation, together with c.3,940m² of external cladding - Thermowood timber cladding and building-integrated solar PV (BIPV) cladding - plus timber window surrounds and finishes.

The hotel has already received multiple awards, including the BREEAM Award for Best New Construction and the edie Net Zero Awards 2025 Green Building Project of the Year. The project was completed in December 2024.

KEY FACTS

MAIN CONTRACTOR
RED Construction

ARCHITECT
ECE Westworks Architects

LOCATION
Babbage Way, Exeter Science Park, EX5 2FN

PROJECT TYPE
New Build, Hotel - 142 bedrooms, four storeys

PACKAGE
Roofing, waterproofing & insulation; c.1,930m² Thermowood timber cladding; c.2,010m² solar PV cladding

PROGRAMME
March 2024 - December 2024

142	c.1,155m²	c.1,930m²	c.2,010m²
BEDROOM HOTEL	HOT MELT MAIN ROOF	TIMBER CLADDING	SOLAR PV CLADDING



Completed hotel
Zeal Hotel, Exeter Science Park

THE CHALLENGES

A heavily serviced roof, a wet start and complex upstands

01

Plant, PV & penetrations

The roof had to support substantial plant, including air handling units, inverters, heat source pumps and solar PV. The steel columns supporting the PV array were each sealed with a hot melt pitch pocket and liquid detail.

02

Weather & co-ordination

Works began in March 2024 during a particularly rainy period, which delayed installation. With multiple trades working on the roof, our dedicated site manager kept the installation on track and made sure the finished roof was protected from following trades.

03

Insulating the upstands

To prevent heat loss, the design required full encapsulation of the upstands. Due to sequencing constraints, Axter developed a hybrid solution combining an inverted roof system with a mix of inverted and warm roof upstands.

PRODUCT SOLUTIONS

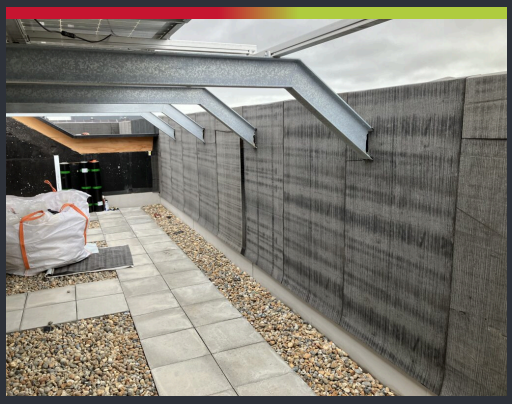
Axter hot melt roofing with timber and solar PV cladding

We wanted a system that would be quick to install and allow a watertight milestone to be reached early - the Axter Wilotekt-Plus hot melt system achieved this. It was selected for its durability and its ability to withstand heavy plant loads and foot traffic.

Axter's support during the design period was crucial in identifying the correct system, quickly building a relationship with our client and the design team. On the main roof the insulation sits above the waterproofing, giving extra protection against damage from plant installations, with a designed U-value of 0.12 W/m²K.

SYSTEMS INSTALLED

- **Axter Wilotekt-Plus inverted roof**
  Hot melt waterproofing, two layers of XPS insulation, ballast and paved walkways
- **Plant, lobby & warm roofs**
  Axter Cityflor to plant and lobby roofs (green roof to the lobby); warm roofs to lift, stair core and 23 pods
- **Thermowood timber cladding**
  c.1,930m² of vertical Thermowood cladding in dark and light finishes
- **Solar PV (BIPV) cladding**
  c.2,010m² of building-integrated solar PV, black matt finish



Ballast & paved walkways
Inverted roof finishes beneath the PV array



Timber & solar PV cladding
Cladding to the main elevation

PROJECT GALLERY

Roofing and cladding at Exeter Science Park



Completed hotel
Exeter Science Park



Timber cladding & window pods
Thermowood cladding



Solar PV over the main roof
PV steelwork sealed through the hot melt



Cladding detail
Timber window surrounds



Main roof
Ballast and paving below the PV



Aerial view
Roof and landscaping

THE RESULT

A complete roof and cladding envelope from one group

The Worx Group delivered the roofing, waterproofing and insulation, and c.3,940m² of timber and solar PV cladding, window surrounds and finishes - completed in December 2024.

The main roof was electronically integrity tested, and Axter's July 2024 quality assurance inspection of the main roof recorded all 16 items as compliant.

PROJECT AT A GLANCE

01 THE CHALLENGE

A heavily serviced roof with plant and PV, a wet start in March 2024 and fully encapsulated upstands.

02 OUR SOLUTION

An Axter-designed Wilotekt-Plus inverted hot melt roof, with Thermowood and solar PV cladding.

03 THE RESULT

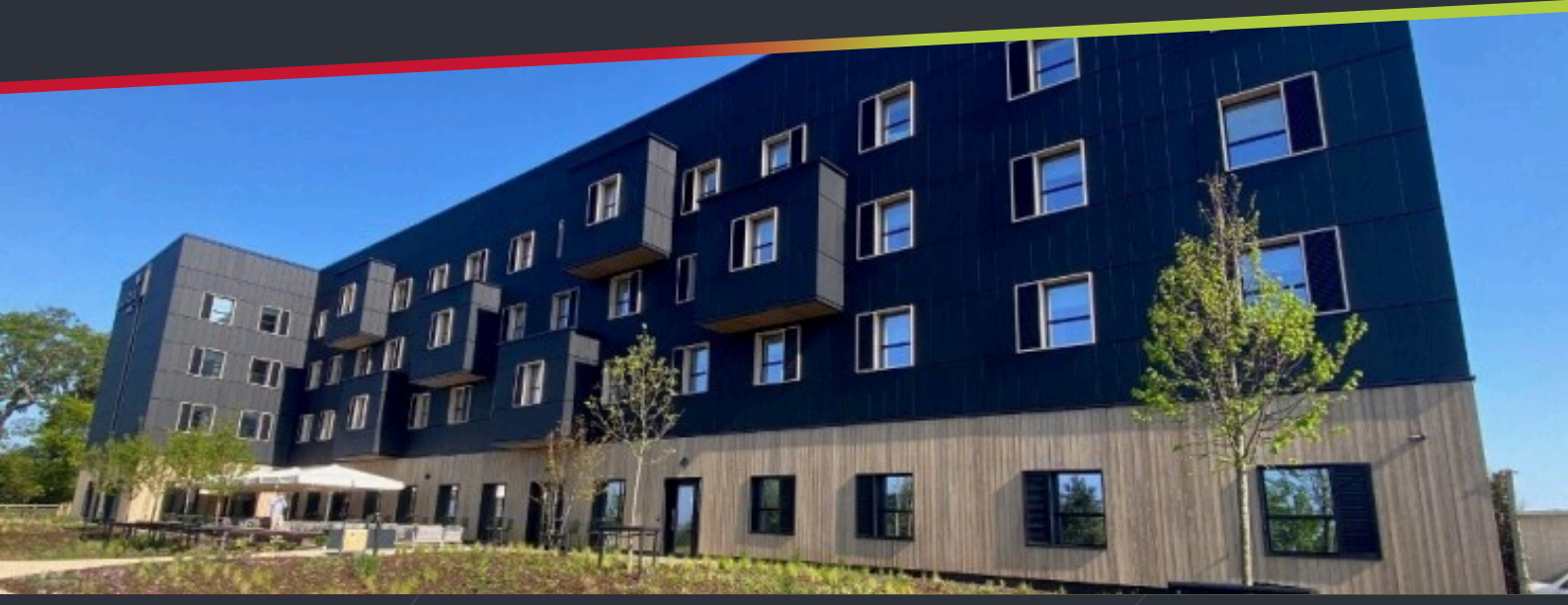
A tested roof with a 35-year guarantee and c.3,940m² of cladding on the UK's first branded net-zero carbon hotel.

AWARD-WINNING

BREEAM Award for Best New Construction and edie Net Zero Awards 2025 Green Building Project of the Year

WHY THE WORX GROUP

- **Bristol based - Nationwide Coverage**
- **Full envelope contractor**
- **Roofs built to last. Guaranteed.**



Completed hotel
Zeal Hotel, Exeter Science Park

ROOFWORX
ROOFING & CLADDING

WOODWORX
CARPENTRY & JOINERY

Bristol based - Nationwide Coverage

- Full envelope contractor
- Roofs built to last. Guaranteed.

Got a project? Let's Talk

enquiries@theworxgroup.co.uk

01173 018152

twg-the-worx-group

www.theworxgroup.uk

TWG
THE WORX GROUP

Part of The Worx Group